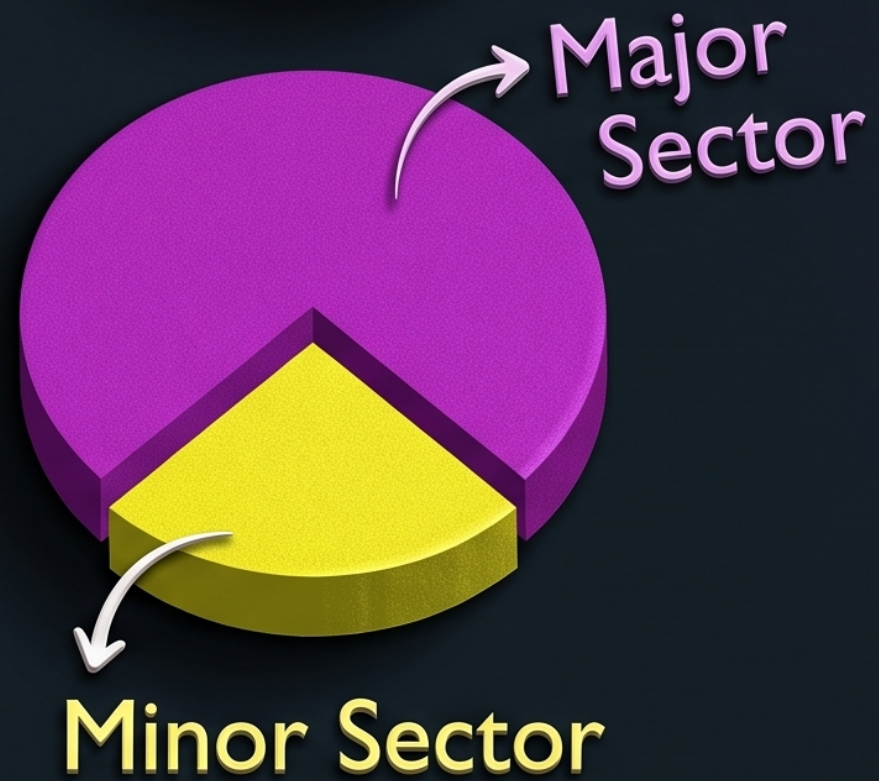
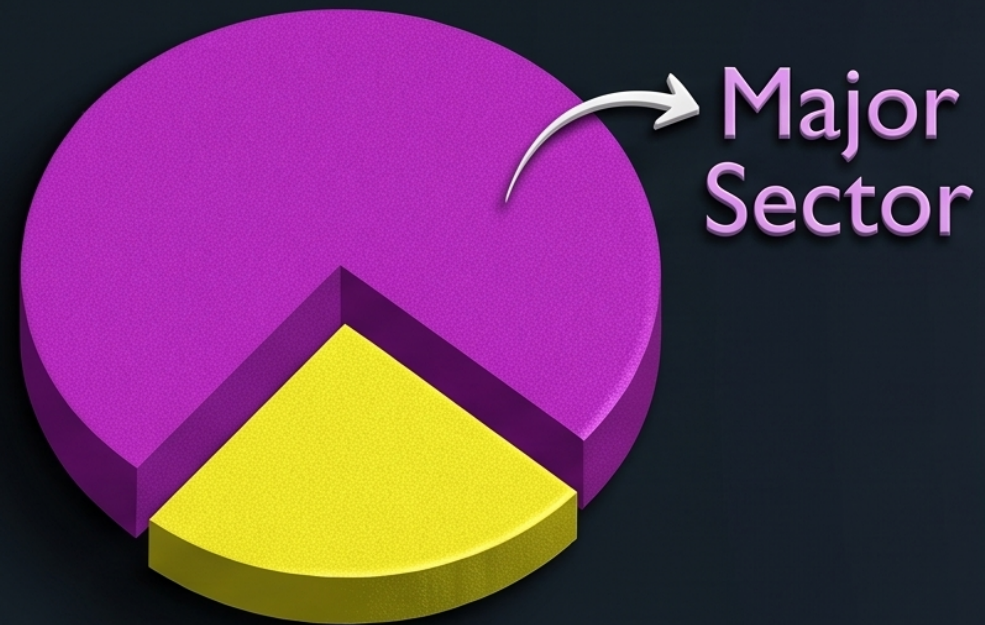
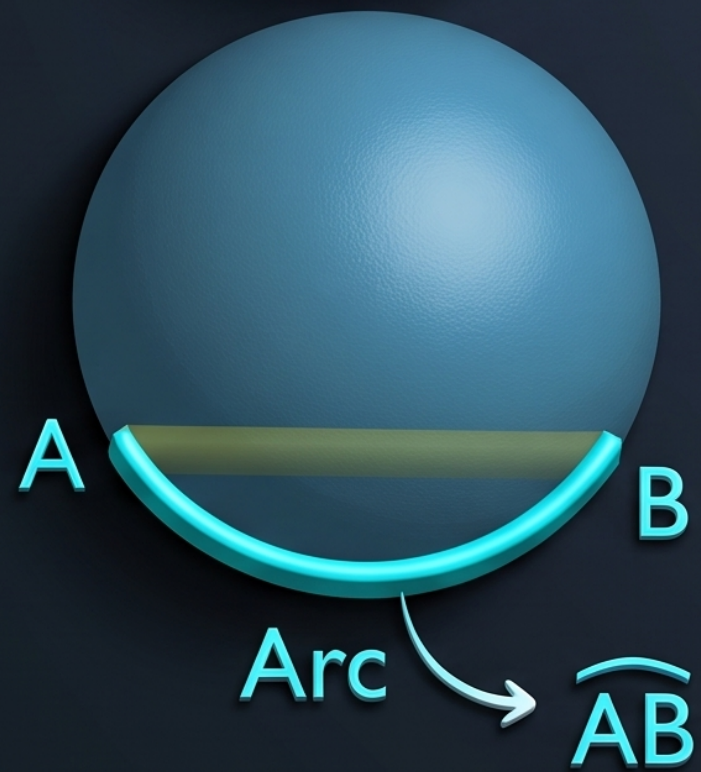


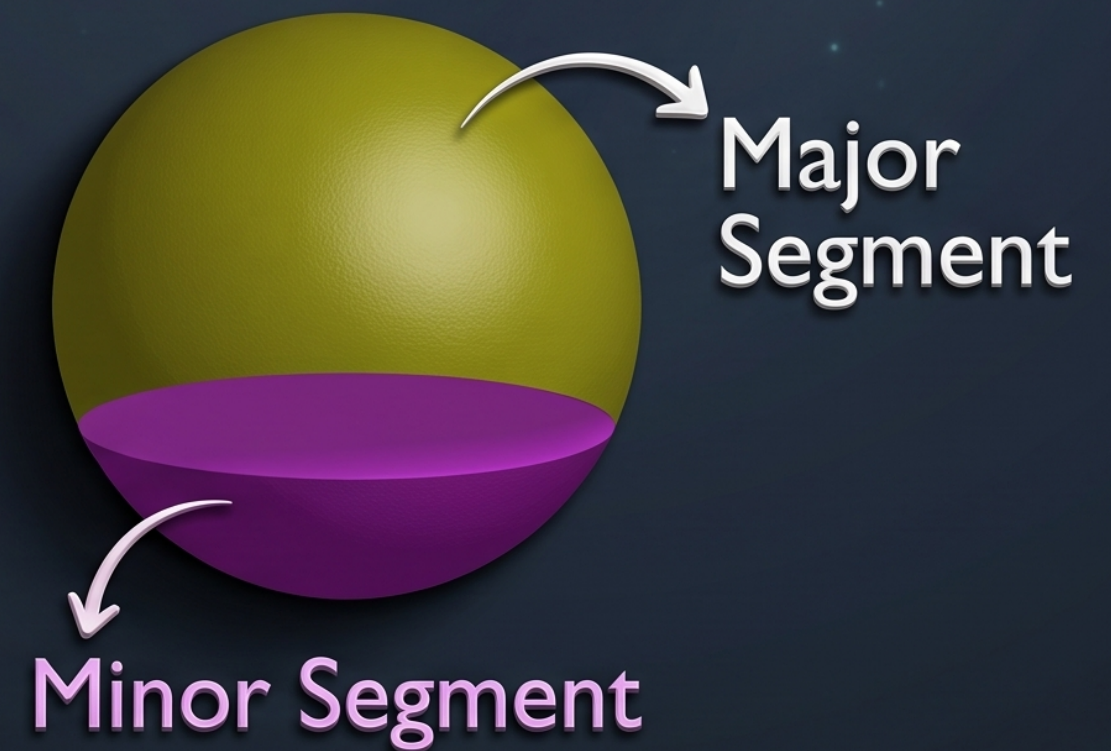
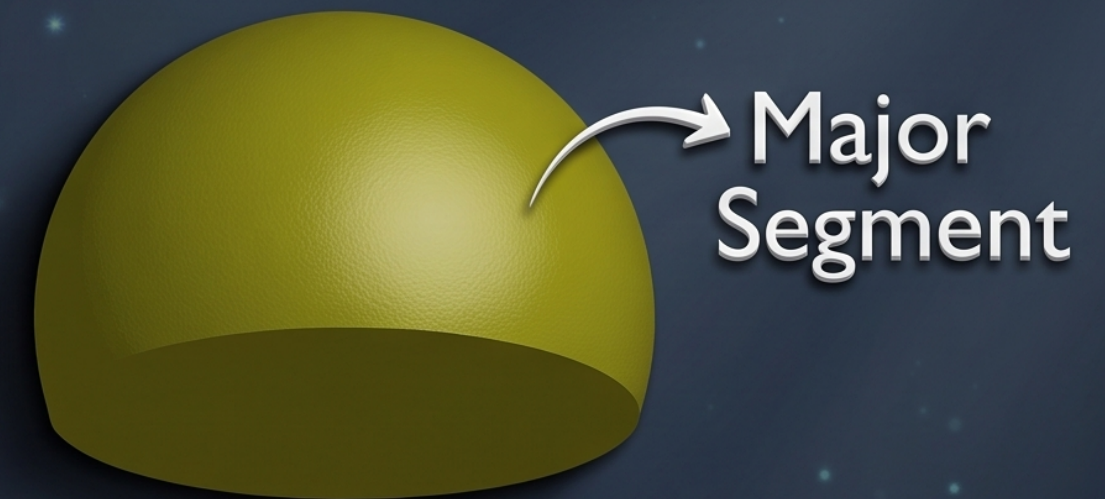
Sector

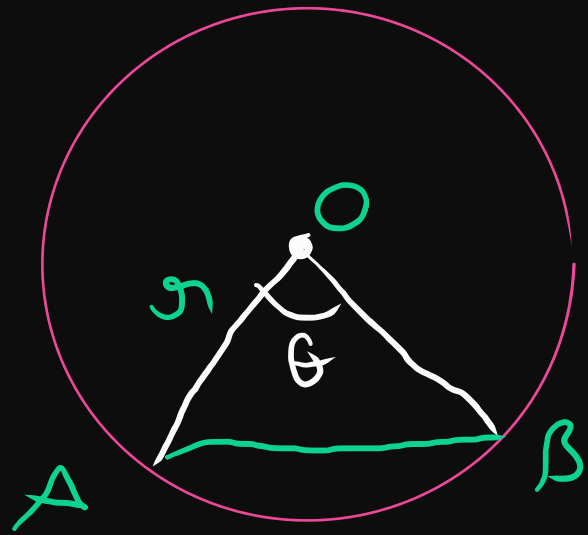


vs



Segment





$$\text{Area of } \triangle AOB \rightarrow \frac{r^2 \sin \theta}{2}$$



$$\triangle \text{ Area} \rightarrow \frac{AB \sin \theta}{2}$$

Area Minor Segment $\rightarrow$ Area Minor Sector $-$ Area $\triangle AOB$

Area Major Segment $\rightarrow$ Area full $-$ Area Minor Segment



$$\theta \rightarrow 30^\circ$$

$$r \rightarrow 7$$

$$(1) A_{Full} \rightarrow 7 \rightarrow 44 \rightarrow \boxed{154}$$

$$(2) A_{minor} \rightarrow \frac{22}{7} \times \frac{7 \times 7}{360} \times 30 \rightarrow \frac{154}{12} = 12.8$$

$$(3) A_{major} \rightarrow \frac{22}{7} \times 7 \times 7 \times \frac{330}{360} \rightarrow \frac{121 \times 7}{6}$$

$$= 154 - 12.8$$

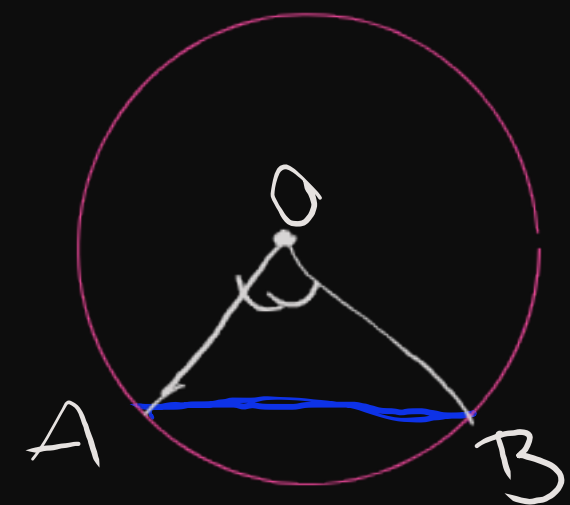
$$(4) C_{full} \rightarrow 44$$

$$(5) C_{minor} \rightarrow \frac{2 \times 22}{360} \times 30 \times 7 = 3.6$$

$$(6) C_{major} \rightarrow \frac{2}{360} \times \frac{22}{7} \times 3 \times 30 \times 7$$

Arc Minor

$$44 - 3.6 = \text{Arc Major}$$



$$(7) \text{ Ar } \triangle AOB \rightarrow \frac{7 \times 7}{2} \sin 30 \rightarrow \frac{49}{4} = 12.2$$

$$(8) \text{ Ar Minor Segment} \rightarrow \text{Ar Minor Sector} - \text{Ar } \triangle AOB$$

$$12.8 - 12.2$$

$$(9) \text{ Ar Major Segment} \rightarrow 154 - 0.6$$

$$= 153.4$$